



LESSON 10.9 IS IT LINEAR?

LESSON INTRODUCTION

MA.8.F.1.1 Given a set of ordered pairs, a table, a graph, or a mapping diagram, determine whether the relationship is a function. Identify the domain and range of the relation.

MA.8.F.1.2 Given a function defined by a graph or an equation, determine whether the function is a linear function. Given an input-output table, determine whether it could represent a linear function.

LEARNING TARGETS

- I can determine whether a function is a linear function given an equation or graph.
- I can determine whether a relation is a function.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.F.1.1 Given an equation or graph that defines a function, classify the function type. Given an input/output table, determine a function type that could represent it.

MA.912.F.1.8 Determine whether a linear, quadratic, or exponential function best models a given real-world situation.

ESSENTIAL QUESTIONS

- How do you determine whether a function is linear given an equation?
- How do you determine whether a function is linear given a graph?
- How do you determine whether a relation is a function?

LESSON NARRATIVE

This lesson builds on the previous lesson of determining whether equations represent a linear function. Students observe equations and their graphs, determining if they are linear and if they are a function. After making observations, students revise their conjectures from the previous day about linear and nonlinear equations. Using their conjecture, students determine which equations from a list are linear functions.

COMPONENT SUMMARY

10.9.1 WARM-UP (~5 MIN.)

Notice and wonder about changes in the U.S. federal minimum wage

10.9.3 BRING IT TOGETHER (5 MIN.)

Complete statements about linear equations and functions

10.9.5 WRAP-UP (~5 MIN.)

Complete statements about linear equations and functions

10.9.2 COLLABORATION (30-35 MIN.)

Determine whether equations and graphs are linear or nonlinear and functions or not functions and make generalizations

10.9.4 YOUR TURN! (5-10 MIN.)

Determine if equations represent linear or nonlinear functions

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms: N/A

MATERIALS

- (Optional) Note cards
- (Optional) Chart paper
- (Optional) Markers

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not recognize linear equations if they are not presented in slope-intercept form, $y = mx + b$.
- Students may mistakenly use horizontal lines instead of vertical lines when conducting the vertical line test.
- Students may mistakenly think absolute value equations are linear.
- Students may confuse the equations of vertical and horizontal lines.
- Students may think that vertical lines are functions.

THINGS TO CONSIDER

- Create an anchor chart and allow students to create reference cards with instructions and examples to aid them in determining if a relation is a function and if a function is linear or nonlinear given equations and graphs.
- Consider having the students explore these graphs and graphs of their own based on their conjectures in Desmos rather than on paper.
- All equations for 10.9.2 Collaboration are prebuilt in Desmos with the graphs turned off.
- Use sliders in Desmos to help students explore the graphs of equations of the format $y = \text{constant}$ versus $x = \text{constant}$. Students need to see that vertical lines are not a linear function because they do not pass the vertical line test.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

10.9.1 Warm-Up involves students noticing what a graph tells them about the relationship between years since 1938 and the federal minimum wage. Then students are asked what they wonder about the graph. Consider incorporating the historical information of the federal minimum wage.

<https://www.dol.gov/agencies/whd/minimum-wage/history/chart>

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD**MA.K12.MTR.4.1 Discussion and Reflection**

10.9.2 Collaboration and 10.9.3 Bring It Together require students to analyze the mathematical thinking of others. Consider elements of MA.K12.MTR.4.1 that facilitate an environment in which students successfully collaborate.

- Establish a culture in which asking questions is a part of learning.
- Reframe and communicate errors as opportunities for learning.
- Require evidence or examples when constructing arguments.

DIFFERENTIATE INSTRUCTION

Encourage discourse while students are completing the 10.9.2 Collaboration. Affirm, recognize, and praise student responses and collaboration that reflect MA.K12.MTR.4.1 in preparation for more precise responses in 10.9.3 Bring It Together.

If resources permit, utilize the digital version of 10.9.2 Collaboration in Desmos to activate a variety of learning opportunities for students.

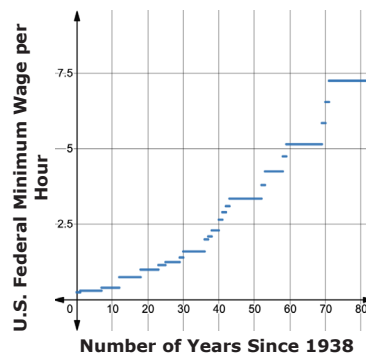
Provide students with the Learning Targets or Essential Questions as additional containers for provided detailed responses in the 10.9.5 Wrap-Up.

10.9.1 WARM-UP: NOTICE AND WONDER**ENRICHMENT**

- What do you know about the minimum wage?
- What do you think the minimum wage should be?

**10.9.1 Warm-Up: Notice and Wonder**

- The graph shows how the value of the U.S. federal hourly minimum wage changed from 1938-2020.



- What do you notice?
Answers will vary.
- What do you wonder?
Answers will vary.



10.9.2 Collaboration: Revising Conjectures about Non-Linear Equations

1. Work with your partner to answer the questions for the description of each equation and its graph.

Equation	Graph	Description
A $y = -2$		Is it linear? ☒ Yes ☐ No Is it a function? ☒ Yes ☐ No
B $y = \frac{1}{2}\sqrt{x} - 1$		Is it linear? ☐ Yes ☒ No Is it a function? ☒ Yes ☐ No
C $\frac{x^3}{2} = y$		Is it linear? ☐ Yes ☒ No Is it a function? ☒ Yes ☐ No
D $y = -x^2 + 6$		Is it linear? ☐ Yes ☒ No Is it a function? ☒ Yes ☐ No

10.9.2 COLLABORATION: REVISING CONJECTURES ABOUT NON-LINEAR EQUATIONS

TEACHER MOVES

Question 1 can be done as a whole group, where the students share whether they think an equation is linear or not based on the previous day’s conjectures and the teacher shows the Desmos activity on a screen for the whole class. Before displaying each graph, the teacher could call on a few students to share whether they think it will be linear or not, sharing their reasoning. This could move to partners and small-group discussion as well, with the teacher revealing each graph after students share their thinking. Students should be revising their conjectures with each new reveal. If using this method, the students should not have their workbooks open. The teacher can then release the students for the rest of this component.

TEACHER MOVES

In order to encourage discourse during this activity, consider requiring that students provide justification to each choice by writing one sentence for each equation. For example, “Equation A is a function because each x-value corresponds to one unique y-value.”

ENRICHMENT

Students must evaluate whether each equation is both linear and a function. This addresses a misconception that linear means a function and vice versa. If students struggle to differentiate the two responses, prompt students to describe the difference between something being linear and being a function.

10.9.2 COLLABORATION: REVISING CONJECTURES ABOUT NON-LINEAR EQUATIONS (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider opportunities to differentiate your approach to the activity based on your classroom resources and the needs of your students.

- Desmos Activity Builder: Utilize digital resources for opportunities to meaningfully manipulate and extend their thinking.
- Gallery Walk: Provide each graph as a poster around the room, with students using markers or pencils to indicate whether each is linear and a function using different symbols.
- Stations: Group equations as prepared Card Sorts in which students must sort before switching to a new set of cards.

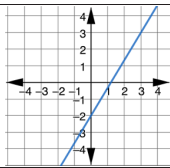
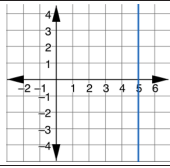
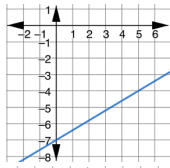
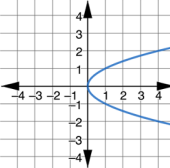
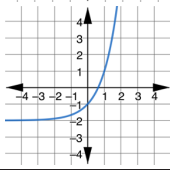
TEACHER MOVES

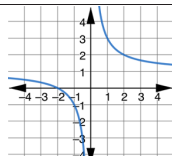
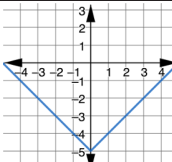
Make note of student responses to Card F to reference for Statement 3 in 10.9.3 Bring It Together. (“An equation is not linear if a constant term is set equal to zero.”)

ENRICHMENT

For students or partners struggling to determine which features inform whether equations are linear or functions, consider the following prompting questions:

- Look at Equations H, I, and J. What do they have in common? (They are all curved, or they are all not straight.) Are they linear?
- Look at those same equations. How are they different? Specifically, think vertically. How is the graph of Equation H different than the graphs of I and J? (It overlaps, or it repeats or reflects.)
- What overlaps or reflects exactly? What is there two of? (The x-values.)
- Say more about what you mean. For example, use the x-value of 4 in H. What happens when $x = 4$? How is that different than I and J? So, how would you describe what to look for when determining what is a function?

E	$5x = 6 + 3y$		Is it linear? ☒ Yes ☐ No Is it a function? ☒ Yes ☐ No
F	$x = 5$		Is it linear? ☒ Yes ☐ No Is it a function? ☐ Yes ☒ No
G	$y = \frac{3}{5}x - 7$		Is it linear? ☒ Yes ☐ No Is it a function? ☒ Yes ☐ No
H	$x = y^2$		Is it linear? ☐ Yes ☒ No Is it a function? ☐ Yes ☒ No
I	$y = 3^x - 2$		Is it linear? ☐ Yes ☒ No Is it a function? ☒ Yes ☐ No

J	$y = \frac{2}{x} + 1$		Is it linear? ☐ Yes ☒ No Is it a function? ☒ Yes ☐ No
K	$y = x - 5$		Is it linear? ☐ Yes ☒ No Is it a function? ☒ Yes ☐ No

Work with your partner to answer the following questions based on your observations of the relations in the table.

- Which equations are **not** functions? Write the equations.
 $x = y^2$; $x = 5$;
- Explain how to use the graph of an equation to determine if the equation represents a function.
By using the vertical line test on a graph, it is possible to determine if the equation is a function. If there is more than one element of the range (y-value) for any element of the domain (x-value), it is not a function.
- What do you notice about the variables x and y in the equations that are linear?
Answers will vary. A possible answer may notice that the both variables are raised to the first power.

10.9.2 COLLABORATION: REVISING CONJECTURES ABOUT NON-LINEAR EQUATIONS (CONTINUED)

TEACHER MOVES

Have students refer to their justifications for each equation when responding to questions 3-5. Consider providing opportunities for students to share their answers while making space for students to agree, disagree, extend, or challenge other responses.

10.9.2 COLLABORATION: REVISING CONJECTURES ABOUT NON-LINEAR EQUATIONS (CONTINUED)

TEACHER MOVES

Use the Stronger and Clearer Each Time routine and ask students to share their answers to question 5 in the Collaboration. Allow the class to make suggestions to create a collective class answer from the shared response.

TEACHER MOVES

For students working in Desmos or if the teacher is displaying it for the whole class, the sliders for the equations $y=-2$ and $x=5$ can be used for more exploration of question 6.

ENRICHMENT

- What is the general form of the equation of a horizontal line?
- What is the general form of the equation of a vertical line?
- Which variable does not change for all ordered pairs on a horizontal line?
- Which variable does not change for all ordered pairs on a vertical line?

5. Write at least one phrase that could be used to complete the statement below.
An equation is **not** linear if either variable is ...
Answers will vary. Possible answers include noticing that the variable is in the denominator, inside a radical, or has an exponent other than 1.
6. The equations of several relations are organized in the table. Complete the statement to describe the graphs of these equations.

Group A	Group B
$y = -2$ $y = 5$ $y = \frac{4}{5}$ $y = -21$	$x = -2$ $x = 5$ $x = \frac{4}{5}$ $x = -21$
The graphs of these equations are ☐ vertical ☒ horizontal lines	The graphs of these equations are ☒ vertical ☐ horizontal lines
because the ☐ x -values ☒ y -values at every point on the graph will be the same.	because the ☒ x -values ☐ y -values at every point on the graph will be the same.

7. Complete the statement.
The equations in Group B ☐ are ☒ are not functions
because ☒ each x -value
☐ at least one x -value corresponds to
☐ exactly one y -value
☒ more than one y -value on the graph.

Alternative answer: ... because at least one x -value corresponds to more than one y -value on the graph.



10.9.3 Bring it Together: Revising Conjectures About Non-Linear Equations

- Ana made several statements after completing the collaboration with her partner. Some of her statements are not true. Cross out any false statements.

Ana's Statements
"An equation is not linear if both variables are raised to a power of 1."
"An equation is not linear if either variable is raised to a power other than 1."
"An equation is not linear if a constant term is set equal to x."
"An equation is not linear if a variable is an exponent."
"An equation is not linear if a variable is in the denominator."
"An equation is not linear if it includes the absolute value or square root of a variable."

- Complete the statements.
 - In a two-variable equation, the equation represents a linear function if both variables are raised to the 1st power, and neither variable is in the

☐ denominator
☐ numerator

 or an exponent.
 - A constant value set equal to

☐ x
☐ y

 can also represent a linear function.
 - An equation represents a linear function if it can be written in the form $y = mx + b$, where m is the slope and b is the y -intercept.

10.9.3 BRING IT TOGETHER: REVISING CONJECTURES ABOUT NON-LINEAR EQUATIONS

QUESTIONING

Challenge students to provide a counterexample for each incorrect statement.

- Statement 1 Counterexample: $y=2x$
- Statement 3 Counterexample: $x=4$

TEACHER MOVES

For the fourth statement, have students read this carefully and clarify with an example. Have the students explain or represent the difference between the phrase "raised to an exponent" and "raised as an exponent."

TEACHER MOVES

Special attention needs to be paid to the concept in 2b. There are sliders in Desmos for this activity that can be used to explore equations that are $y = \text{constant}$ versus $x = \text{constant}$. Students need to understand that while equations in the form $x = \text{constant}$ are a line, they are not a linear function. This is an opportunity to spiral the definition of a function as well.

TEACHER MOVES

Question 2c is included because it is in the BEST document when defining linear functions. It's important that students don't focus solely on this, as they will see linear equations written in many different forms and should not have to rewrite them to determine if they are linear. More emphasis should be placed on questions 2a and 2b here. Question 2c reviews concepts introduced in Unit 10.

10.9.4 YOUR TURN!

TEACHER MOVES

Before releasing students to do Your Turn!, revisit the Wrap-Up from Lesson 8. Allow students to work with partners to revise their answers or refer to the class data collected.

ENRICHMENT

Why is $y = 3,278$ considered a function? Are all 'straight' lines functions? Give an example or counterexample to support your claim.

TEACHER MOVES

Turn and Talk

Frame question 2 as an opportunity to practice mathematical literacy, thinking, and reasoning. Have students find a partner who chose a different equation and explain their rationale. Have students agree or disagree and share any findings.



10.9.4 Your Turn!

- Complete the table.

Equation	Linear Function	Not a Linear Function
$y = 3x^2 + 5x$	☐	☒
$y = 4 + \frac{1}{6}x$	☒	☐
$y = 3,278$	☒	☐
$y = \sqrt{2x + 9}$	☐	☒
$y = x^4 - 7$	☐	☒
$x = -24$	☐	☒
$-4x = 9y + 13$	☒	☐
$y = 5^{x+1}$	☐	☒

- Choose one equation from the table and explain why it is not a linear function.
Answers will vary. Possible responses include:
 $y = 3x^2 + 5x$; In one of the terms, the x is raised to the second power.
 $y = \sqrt{2x + 9}$; The x -variable is under a radical symbol (or square root)
 $y = x^4 - 7$; The x -variable is raised to the 4th power.
 $x = -24$; This is a vertical line, which is not a function.
 $y = 5^{x+1}$; The x -variable is located in the exponent.
- Choose one equation from the table and explain why it is a linear function.
Answers will vary. Possible responses include:
 $y = 4 + \frac{1}{6}x$ and $-4x = 9y + 13$; Both the x - and y -variables are raised to the first power and located in the numerator.
 $y = 3,278$; This is a horizontal line.

**10.9.5 Wrap-Up: Reflection**

Complete each statement as you reflect on today's learning.

1. When someone asks me what I did in math today, I can say ...
2. The thing that made the most sense to me today was ...
3. Something that I still do not understand is ...

Answers will vary.

10.9.5 WRAP-UP: REFLECTION**DIFFERENTIATE INSTRUCTION**

Consider using the responses from the Wrap-Up to inform small groups or partner work for Guided Practice in Lesson 10. Consider providing the lesson Learning Targets as sentence stems for students.